

Climate Chain: From Gas to Warming

GROW • Science • 40 minutes

I can explain how extra greenhouse gases raise Earth's average temperature and connect warming to wider climate effects.

Prepare before pupils arrive

Setup: Prepare a weather card, a many-year climate strip, energy-route arrows and the five genuine causal-chain cards. Keep visuals calm and factual.

Safety: Do not use a bottle-and-lamp analogy, personal guilt prompts or catastrophic footage. Allow a seated director route.

Equipment: Cause, energy and effect cards; arrow strip; pencil; optional calm digital card route.

Safety: Use calm, solution-aware language. Do not ask for family carbon data or use frightening imagery.

Fallback: Order printed cards or direct an adult to move them. A scribed exact response is equivalent evidence.

Access and participation

Offer reading aloud, pointing, signing, quiet thinking, a no-touch route and exact scribing where these support the learner. Keep the same subject decision. In shared work, let each learner commit before feedback; use a partner as card mover or checker, then swap. Avoid public speed rankings.

Source lesson curriculum link: Describe causes and effects of global warming and how to reduce our impact.

40-minute teaching sequence

Stage / total time	Teaching move	Adult support / response
Arrival · 4 min	State one difference between weather and climate. Recall why repeated measurements are useful.	Wait, regulate and retrieve through the lightest workable route. If recall is inaccessible, point to one retrieval sentence; do not teach today's answer.
Starter · 4 min	Think first. Point, say, sign or write your choice.	Protect curiosity: receive every first idea without judgement. Ask 'What makes you think that?' and preserve the prediction for later evidence.

Stage / total time	Teaching move	Adult support / response
I do · 5 min	climate — weather pattern over time greenhouse gas — gas that affects heat flow average — one value for a set Sunlight passes through the atmosphere and transfers energy to Earth's surface. The warm surface gives out infrared energy.	Let the teacher/model carry the explanation; pupils watch, point or track. Use a visual cue only if access is the barrier, then fade it.
We do · 5 min	Commit to an answer. Show the evidence you used.	Scan every response mode. Do not infer understanding from handwriting or confidence. Secure: transfer. Mixed: contrast. Misconception: counterexample then a fresh question.
I do 2 · 4 min	Human activities such as burning coal, oil and gas, plus deforestation, add greenhouse gases. A stronger greenhouse effect raises the global average; it does not make every day or place equally hot.	Model one complete evidence-to-explanation sentence, then pause. Keep the science words; change density, modality or adult role before changing the goal.
We do 2 · 4 min	Order one genuine five-link chain from a human activity to one likely effect. Say why each link follows the one before it.	Protect pupil operation and thinking; support handling only where needed. Ask what changed and what stayed the same before supplying a scientific word.
Independent · 10 min	Create the five-link chain activity → more greenhouse gas → slower net energy loss → higher global average → one likely effect, then explain it aloud, by pointing or in writing. Record: One ordered causal chain, one explained effect and one honest model limitation.	Protect independence. Accept equivalent evidence routes and scribe exact meaning. Prompt only as much as needed; fade as soon as the pupil continues.
Exit · 4 min	What changes in the energy route when extra greenhouse gas is added? Point to, read or explain the evidence you made.	Genuinely receive one final response and name back the Science heard or seen. Choose one visible next move. The pupil writes what they said and what changed.

The timing belongs to the whole stage. Several PowerPoint slides may share it; do not restart that time on every slide. Full source-stage text is in the PowerPoint speaker notes and original HTML.

Answers, evidence and feedback

Hinge answer: Climate is the usual pattern over many years; one cold day is weather.

Yes. Weather can vary from day to day while the long-term global average rises.

Misconception repair

Repaired. Climate evidence uses long-term patterns from many measurements; cold weather can still occur.

Guided checkpoint

Yes. Weather can vary from day to day while the long-term global average rises. Compare a short event with a pattern measured over many years.

Burning fossil fuels

1 · HUMAN ACTIVITY. This is the first link: burning fossil fuels is a human activity that releases carbon dioxide. Start with the human activity, before the amount of gas or the energy route changes.

More greenhouse gas in the atmosphere

2 · MORE GREENHOUSE GAS. This is the second link: the activity raises the amount of greenhouse gas in the atmosphere. Place the change in atmospheric gas after the activity and before the energy change.

Net energy loss slows

3 · ENERGY CHANGE. This is the energy mechanism: extra greenhouse gas slows the net loss of energy to space. Place the infrared-energy change after the gas increases and before the average temperature rises.

Global average temperature rises

4 · GLOBAL WARMING. This is global warming: the long-term global average temperature rises. Place the temperature response after net energy loss slows and before a wider climate effect.

Land ice melts

5 · LIKELY EFFECT. This is an effect linked to warming. Melting land ice adds water to the ocean. Place the outcome after the cause and energy mechanism.

Model limitation

The arrows show direction, not real amounts, times or the full climate system. Greenhouse gases are not a solid lid and do not create energy.

Independent evidence

One ordered causal chain, one explained effect and one honest model limitation.

Supported: Order the five icon-backed links and complete: More greenhouse gas slows _____. The higher global average can lead to _____. Two cards are pre-positioned; use pointing, speech, symbol choice or exact adult scribing.

Standard: Draw or order the full chain and explain one effect using because. Use the three headings Cause, Energy mechanism and Effect, then remove the sentence frame.

Stretch: Contrast the natural and enhanced greenhouse effects, then explain why one weather event is weak evidence. Add one model limitation and use changed risk rather than guaranteed outcome.

Record the teaching response

What the learner actually showed	Next teaching action
Secure explanation with evidence	Reduce content prompting; offer another example or a limitation.
Partly correct / mixed	Point to the deciding evidence and invite a revision.
Access barrier	Change the reading, sensory or response format; keep the subject decision.
Missing source or observation	Keep the gap visible. Name the source or authorised check needed.

Safety and evidence boundary

Use the centre's authorised practical or fieldwork method and local risk assessment. Supplied sample data, fictional place models, card feedback and rehearsals are teaching material. They are not the learner's practical observations or proof of a qualification outcome. For portfolio use, check the exact registered unit/version, accepted evidence and centre process.

Sources and companion notes

Primary classroom source: [SCI_G_W11A_Global_Warming_Explore.html](#)

The uploaded lesson is included unchanged. Text and teaching steps in the PowerPoint are editable; source diagrams are placed images. Word booklets are editable. Static PDFs cannot reproduce HTML controls; use the paper cards/tables or open the original HTML.

Verification scope: matched to the uploaded title, pathway, objective, stage sequence and 40-minute timing. Embedded workbook references are retained as source locators; this is not a new line-by-line audit of every scheme-of-work row or a centre assessment sign-off.